Teorie 2

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1) Disc: $x^2+y^2<1$; $x^2+y^2-2ay<0$; ext. disc: ... $>R^2$

2) Interiorul unei elipse: $8x^2+y^2<1$; ext: ... >1

3) Sferă: $x^2+y^2+z^2=1$ sau $x^2+y^2+z^2\le 1$; $z>0 \Rightarrow$ semisferă

4) Cilindru: $x^2+y^2=a^2$ (în 3D)

5) Cerc: $x^2+y^2=a^2$ (în xoy)

6) Elipsă: $\frac{x^2}{25}+\frac{y^2}{9}=1$

7) Hiperbolă echilateră: $xy=2$

8) Dreaptă: $y=2x$

9) Semidisc: $y\ge 0$

10) Con: $z=\sqrt{x^2+y^2}$ (vârf în origine)

11) Paraboloid: $x^2+y^2=z$

12) Elipsoid: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

13) Ecuația unei drepte

$y=mx+n$

$y<mx+n$ domeniul situat sub dr.

$y>mx+n$ —||— deasupra —||—

Parametrizarea elipsei: $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$$\begin{cases} x=a\cos t \\ y=b\cdot\cos t \end{cases},\ t\in[0,2\pi]$$

Coordonate polare generalizate: $\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1$

$$\begin{cases} x=a\rho\cos\theta \\ y=b\rho\sin\theta \end{cases},\ |J|=ab\rho \qquad \rho\in[0,R]$$

Coordonate sferice gen.: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1$ $\qquad \theta\in[0,\pi]$

$$\begin{cases} x=a\rho\sin\theta\cdot\cos\varphi \\ y=b\rho\sin\theta\cdot\sin\varphi \\ z=c\rho\cos\theta \end{cases},\ |J|=abc\rho^2\sin\theta \quad \varphi\in[0,2\pi]$$

Parametrizarea canonică a sferei:

$$\begin{cases} x(\theta,\varphi)=R\sin\theta\cos\varphi \\ y(\theta,\varphi)=R\sin\theta\sin\varphi \\ z(\theta,\varphi)=R\cos\theta \end{cases},\ |J|=R^2\sin\theta$$

Volumul corpului delim. de domeniul D și $f(x,y)$:

$\iint_D f(x,y)\,dxdy$

Volumul:

$Vol(\Omega)=\iiint_\Omega dxdydz$

- trecem la coordonate sferice

Aria:

$Aria(D)=\iint_D dxdy$

- trecem la coor. polare

Vectorii tangenți

$\psi: D\to\mathbb{R}^3$

$\frac{\partial\psi}{\partial u}, \frac{\partial\psi}{\partial v}$ vect. tg

$\psi: D\to\mathbb{R}^3,\ \psi(u,v)=(x,y,z)$

$\frac{\partial\psi}{\partial u}=\left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)$

$\frac{\partial\psi}{\partial v}=\left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)$

$x\in[a,b],\ \varphi_1(x)\le y\le\varphi_2(x)$

$\iint_D f(x,y)\,dxdy=\int_a^b\left(\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,dy\right)dx$

$f(x,y)=1$

$aria(D)=\int_a^b(\varphi_2(x)-\varphi_1(x))\,dx$

Integrala de suprafață de speța I

$$\int_\Sigma f\,d\sigma=\iint_D f(\underset{\downarrow}{\psi(u,v)})\left\|\frac{\partial\psi}{\partial u}\times\frac{\partial\psi}{\partial v}\right\|dudv$$

$$\int_\Sigma F(x,y,z)\,d\sigma=\iint_D F(x,y,f(x,y))\sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dxdy$$

"z" în funcție de x și y

$\|\bar N(x,y)\|=\sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}$

$aria=\iint_D\sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dxdy$

Normala la suprafață

$\bar N=\frac{\partial\psi}{\partial u}\times\frac{\partial\psi}{\partial v}$

$\bar\nu=\frac{\bar N(u,v)}{\|\bar N(u,v)\|}$ versor

Fluxul vectorial / Integrala de suprafață de speța a II-a

$F_\Sigma(\bar v)=\int_\Sigma\bar v\cdot\bar N\,d\bar r$

$\bar r=x\bar i+y\bar j+f(x,y)\bar k$; $z=f(x,y)$

$\bar N=\pm\frac{\bar r_x+\bar r_y}{\|\bar r_x\times\bar r_y\|}$ (în funcție de sens)

$F_\Sigma(\bar v)=\pm\iint_D\underbrace{\bar v\cdot\bar r_x\times\bar r_y}_{\text{determinant}}\,dxdy$

elem $\bar v$
elem r_x
elem r_y

Câmp de gradienți

$\Leftrightarrow\int_{S(0,R)}\bar v\,d\bar r=0$

$\bar v=P\bar i+Q\bar j+R\bar k$ câmp vectorial

$C=\int_\gamma\bar v\,d\bar r=\int P\,dx+Q\,dy+R\,dz$

circulația c.v. $\bar v$ de-a lungul curbei γ

$L(\gamma)=\int_\gamma ds$ – lungimea curbei γ

$F_S(\bar v)=\int_S\bar v\cdot\bar n\,d\bar r=\iiint_\Omega\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dxdydz$

fluxul

$ds=\sqrt{\left(\frac{\partial x}{\partial t}\right)^2+\left(\frac{\partial y}{\partial t}\right)^2+\left(\frac{\partial z}{\partial t}\right)^2}$ (parametrizăm)

Green-Riemann $\quad\bar v:(P,Q)$, Dacă $\frac{\partial P}{\partial y}-\frac{\partial Q}{\partial x}$ sau $\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}=$ const

$$\int_\gamma\omega=\iint_K P\,dx+Q\,dy=\iint_K\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy$$

Gauss-Ostrogradski

divergența $div(\bar v)=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}$

$F_\Sigma(\bar v)=\int_K\bar v\,\bar N\,d\bar r=\iiint_K div(\bar v)\,dxdydz$

Stokes

$\int_\gamma\bar v\,d\bar r=\int_\Sigma rot\,\bar v\cdot\bar N\,d\bar v$ circulația

$\iint_D rot\,\bar v\cdot r_x\times r_y\,dxdy$

$$Rot(\bar v)=\begin{vmatrix}\bar i & \bar j & \bar k\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ P & Q & R\end{vmatrix}$$

Lucru mecanic (circulația)

$\int_\gamma\bar v\,d\bar r=\int_a^b P\,dx+Q\,dy+R\,dz$

Funcțiile lui Euler Γ, B (Gama și Beta)

Def $\Gamma:(0,\infty)\to\mathbb{R}$

$\Gamma(\alpha)=\int_0^\infty x^{\alpha-1}\cdot e^{-x}\,dx$

Prop $\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$

Obs $\Gamma(\ =\int_0^\infty e^{-x}\,dx=-e^{-x}\Big|_0^\infty=1$

$\Gamma(2)=\Gamma(1+1)=1\cdot\Gamma(1)=1$

$\Gamma(3)=\Gamma(2+1)=2\Gamma(2)=2\cdot1=2$

$\Gamma(4)=\Gamma(3+1)=3\Gamma(3)=3\cdot2\cdot1=6$

$\Gamma(n)=(n-1)!$

Proprietăți:

Funcția Beta

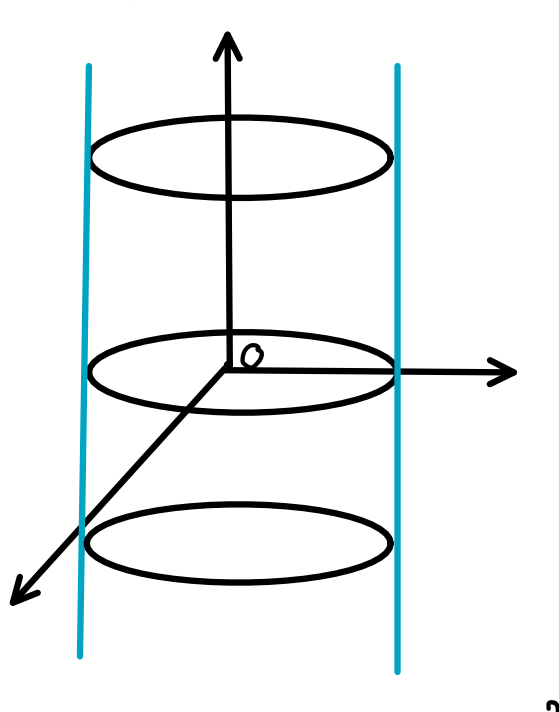
$B:(0,\infty)\times(0,\infty)\to\mathbb{R}$

$B(p,q)=\int_0^1 x^{p-1}(1-x)^{q-1}\,dx$

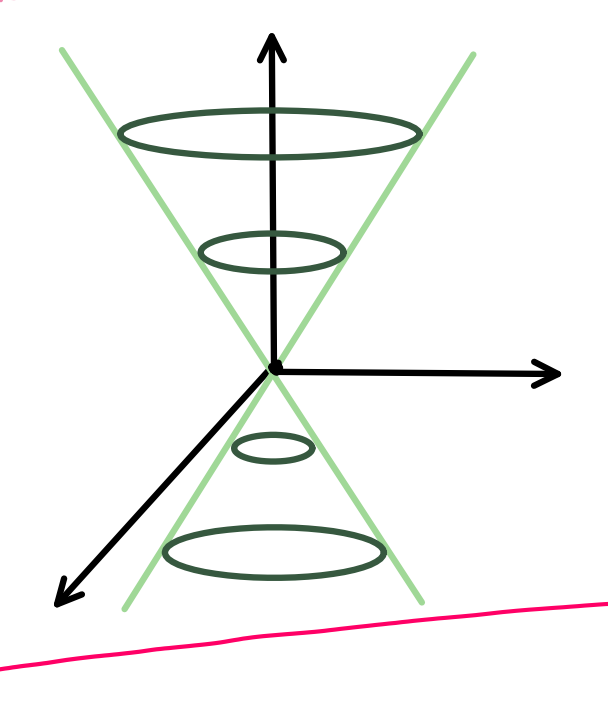
1) $\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$
2) $\Gamma(n)=(n-1)!,\ \forall n\in\mathbb{N}^*$
3) $B(p,q)=\int_0^\infty\frac{y^{p-1}}{(1+y)^{p+q}}\,dy$
4) $\boxed{B(p,q)=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}}$
5) $\alpha\in(0,1)$ $\Gamma(\alpha)\cdot\Gamma(1-\alpha)=\frac{\pi}{\sin(\alpha\pi)}$

Desene

1) Cilindru



2) Con



$z^2=x^2+y^2$

$z=0\to x=y=0$

$x=0\Rightarrow z^2=y^2$

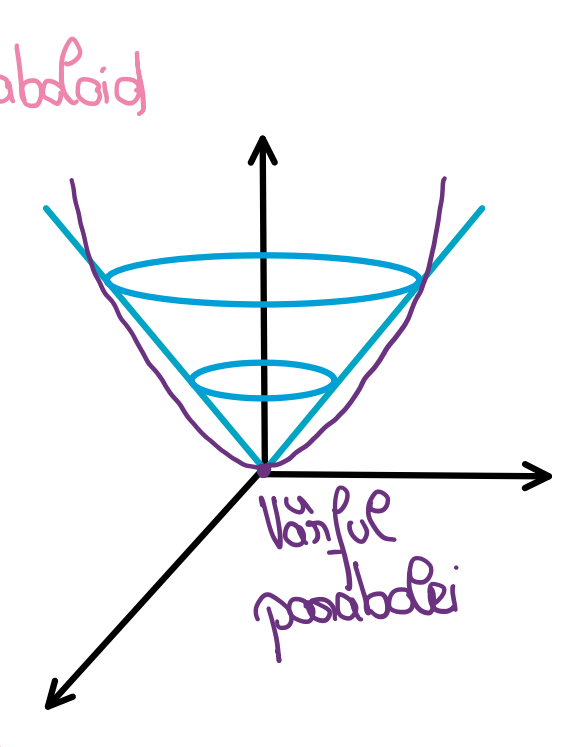
$(z-y)(z+y)=0$

$z=y$ și $z=-y$

Coordonate cilindrice

$$\begin{cases} x=\rho\cos\theta \\ y=\rho\sin\theta \\ z=z \end{cases},\ |J|=\rho$$

3) Paraboloid



$z=y^2+x^2$

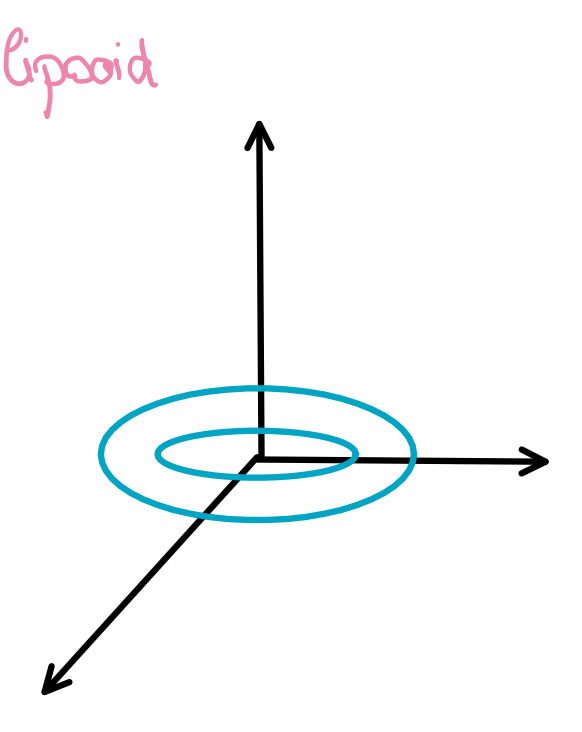
$z=0\to x=y=0$

$z=1\to x^2+y^2=1$

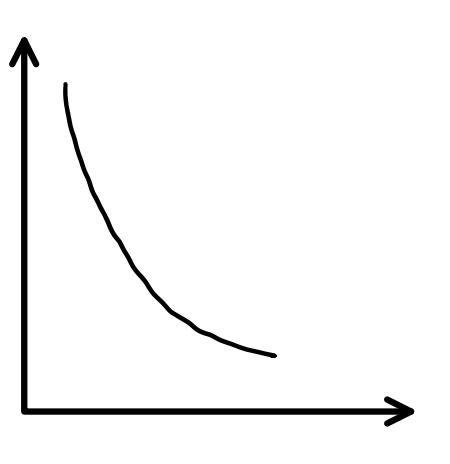
$z=2\to x^2+y^2=2$

$x=0\Rightarrow y^2=z$

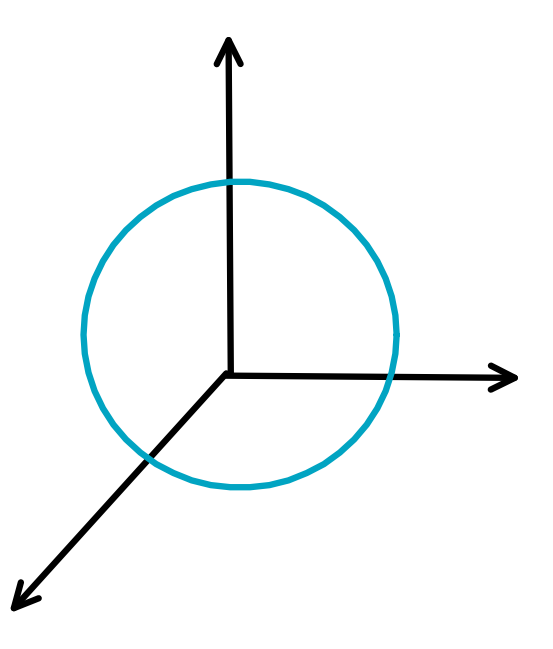
4) Elipsoid



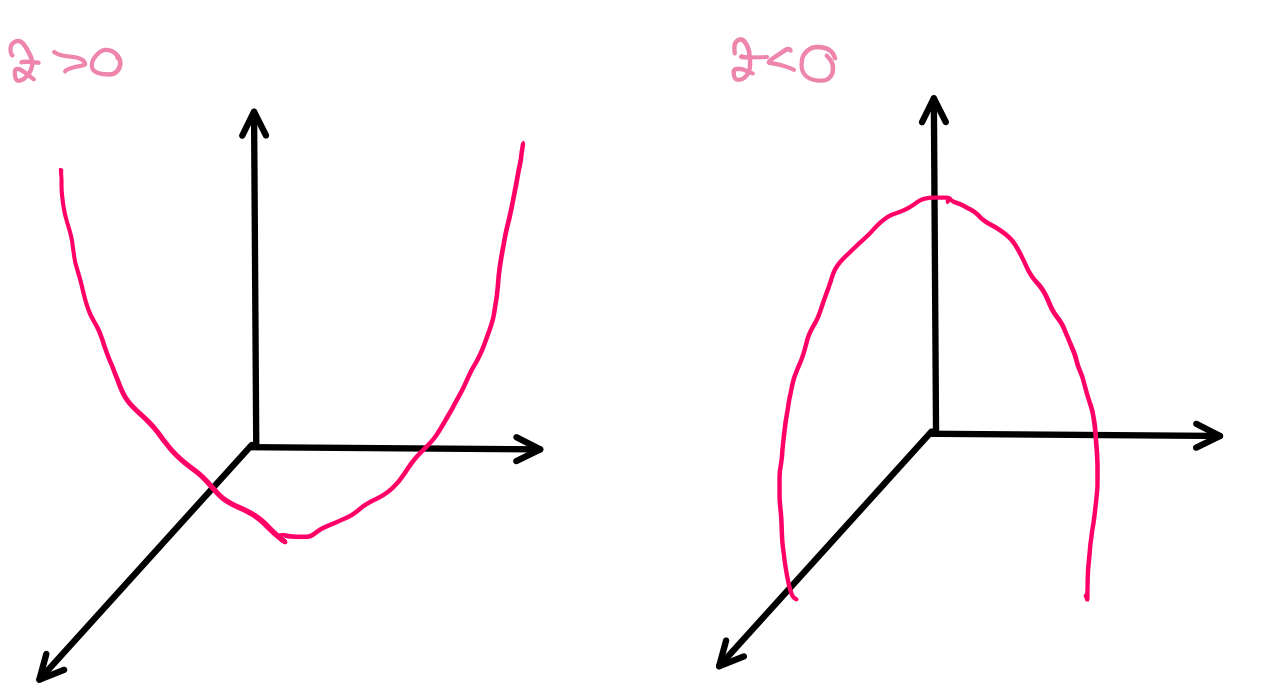
5) Hiperbolă echilateră



6) Sferă

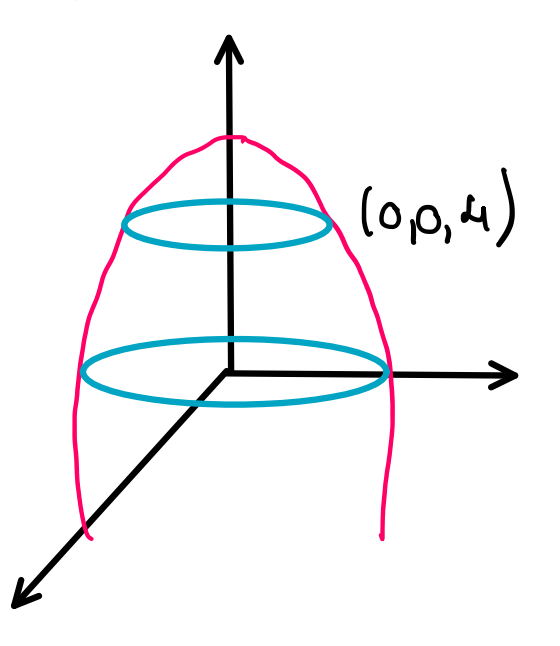


7) Paraboloid

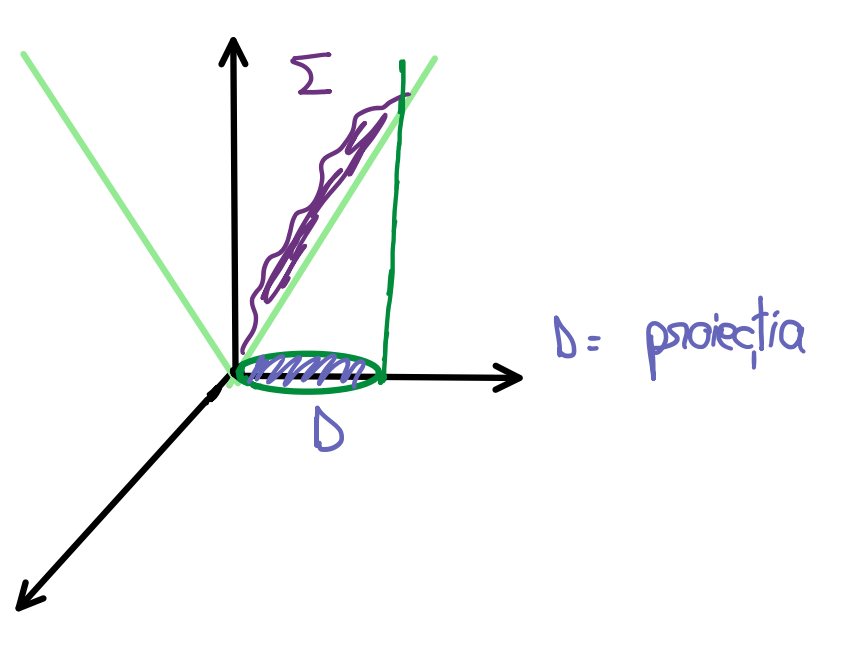


ex: $4-z=x^2+y^2$

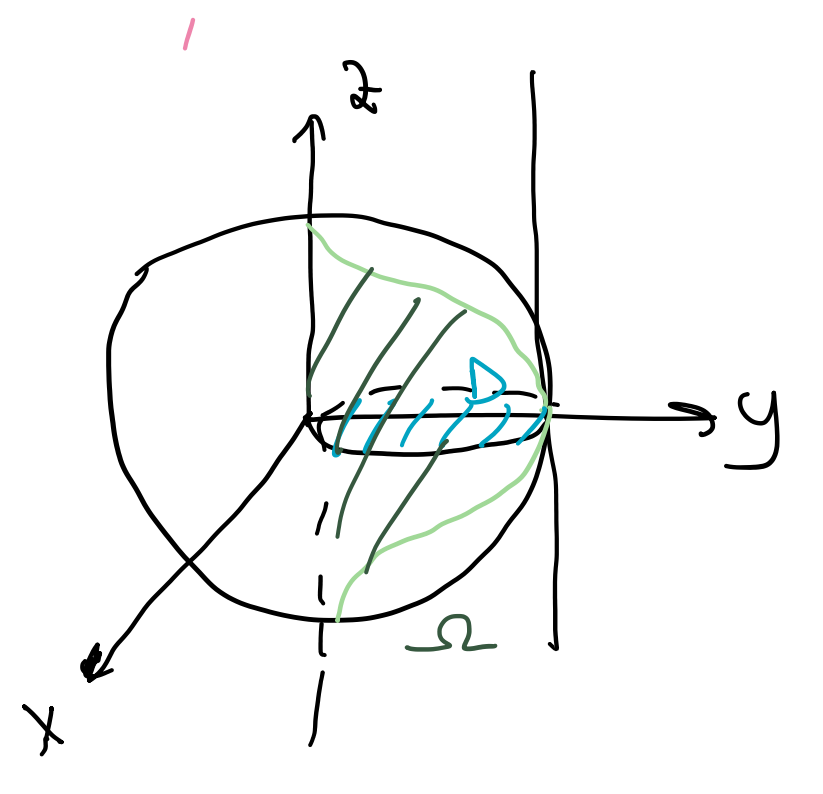
$R=4$



8) Con cu cilindru



9) Disc și



Volum sferă: $\frac{4\pi R^3}{3}$

Volum con: $\frac{\pi R^2 h}{3}$

Arie cerc: πR^2

Volum cilindru: $\pi R^2 h$

Fluxul câmpului electric:

$\|F\|=\sqrt{x^2+y^2+z^2}$

$r=\|F\|$

$r(x,y,z)=\sqrt{x^2+y^2+z^2}$

$\bar E=\frac{q}{4\pi}\cdot\frac{\bar r}{r^3}$

Câmp electric: $\frac{1}{r^2}$

Exercițiu

$div(E)=0$

Fie Σ o suprafață oarecare, unde $0\notin\Sigma$

$K=Int\,\Sigma$ mărginită

$E=\partial K$

$F_\Sigma(E)=?$

I $0\notin Int\,\Sigma\overset{G-O}{=}F_\Sigma(\bar E)=0$

II $0\in Int\,\Sigma$

Fie $S(0,R)\subset Int\,\Sigma$

$K=Int\,\Sigma$

$K'=K\setminus Int\,S(0,R)$

$\partial K'=\Sigma\cup S(0,R)$

$\downarrow$ $\bar n$ ext Σ $\qquad\downarrow$ $\bar n$ int la sferă

$F_{\partial K'}(\bar E)\overset{G-O}{=}0$

$F_\Sigma(\bar E)+F_{S(0,R)}(\bar E)=0$

$F_\Sigma(\bar E)=-F_{S(0,R)}(\bar E)=F_{S(0,R)}(\bar E)\to$ cu dif. fluxurilor (rez. este q!) legea lui Gauss

$\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}\Rightarrow\omega$ – închisă ; $\mathbb{R}^2$ – stelată $\Big|\Rightarrow\omega$ – exactă

$\downarrow$ mulțimea convexă

$\overset{TP}{\Rightarrow}$ integrala independentă de drum

$\to\int_\gamma\omega=\int_{[AB]}\omega$

$\bar v$ câmp de gradienți $\Leftrightarrow\int_\gamma\bar v\,d\bar r=0$ $\forall\gamma$ curbă închisă

Serii Fourier

$$f(x)=\frac{a_0}{2}+\sum_{n\ge1}\left(a_n\cos(nx)+b_n\sin(xn)\right)$$

$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\,dx$

$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx,\ \forall n\ge1$

$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nx\,dx,\ \forall n\ge1$

! intervalele trb. să fie de per. 2π

$f(x)$ – pară $\Rightarrow b_n=0,\ \forall n\ge0$

$f(x)$ – impară $\Rightarrow a_n=0,\ \forall n\ge0$

Identitatea lui Parseval

$$\frac{a_0^2}{2}+\sum_{n\ge1}(a_n^2+b_n^2)=\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx$$

Distanța Pompeiu Hausdorff

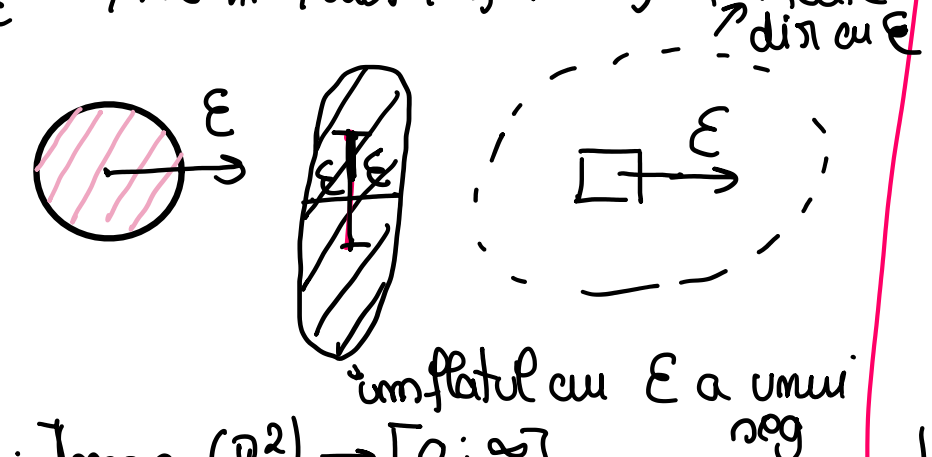
K – imagine

$P\in\mathbb{R}^2$ (punct)

$dist(P,K)=\min\{d(P,M)\mid M\in K\}$

Fie $K\in Imag$, fie $\varepsilon<0$

$K_\varepsilon=\{M\in\mathbb{R}^2\mid dist(M,K)\le\varepsilon\}$ o dilatare în toate dir. cu ε



umflatul cu ε a unui seg

$h: Imag(\mathbb{R}^2)\to[0;\infty]$

$K, L\in Imag(\mathbb{R}^2)$

$h(K,L)=\underset{\min}{\inf}\left\{\varepsilon<0\ \middle|\ \begin{matrix}K\le L_\varepsilon\\ L\le K_\varepsilon\end{matrix}\right\}$

$(Imag, h)$ = spațiu metric complet

1) Translația de vector (a,b)

$T_{(a,b)}(x,y)=(x+a,y+b)$

nu e contracție

2) Rotații de unghi α

$$R_\alpha(x,y)=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}$$

3) Omotetie de factor h

$O_h(x,y)=(hx,hy)$

Dacă $h\in(0,1)$, O_h – contracție

Operator Hutchinson:

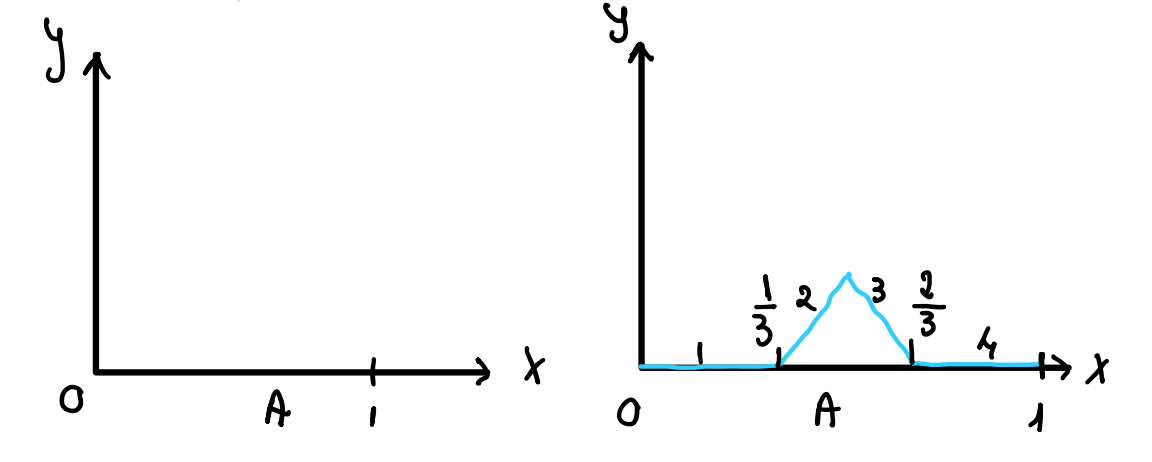
Fie $\omega_1,\omega_2,\dots,\omega_k$ contracție în plan

Fie $H: Imag(\mathbb{R}^2)\to Imag(\mathbb{R}^2)$

$H(A)=\omega_1(A)\cup\omega_2(A)\cup\dots\cup\omega_k(A)$

H contracție

Curba lui Koch



$\omega_1(x,y)=\left(\frac13x,\frac13y\right)=O_{\frac13}(x,y)$

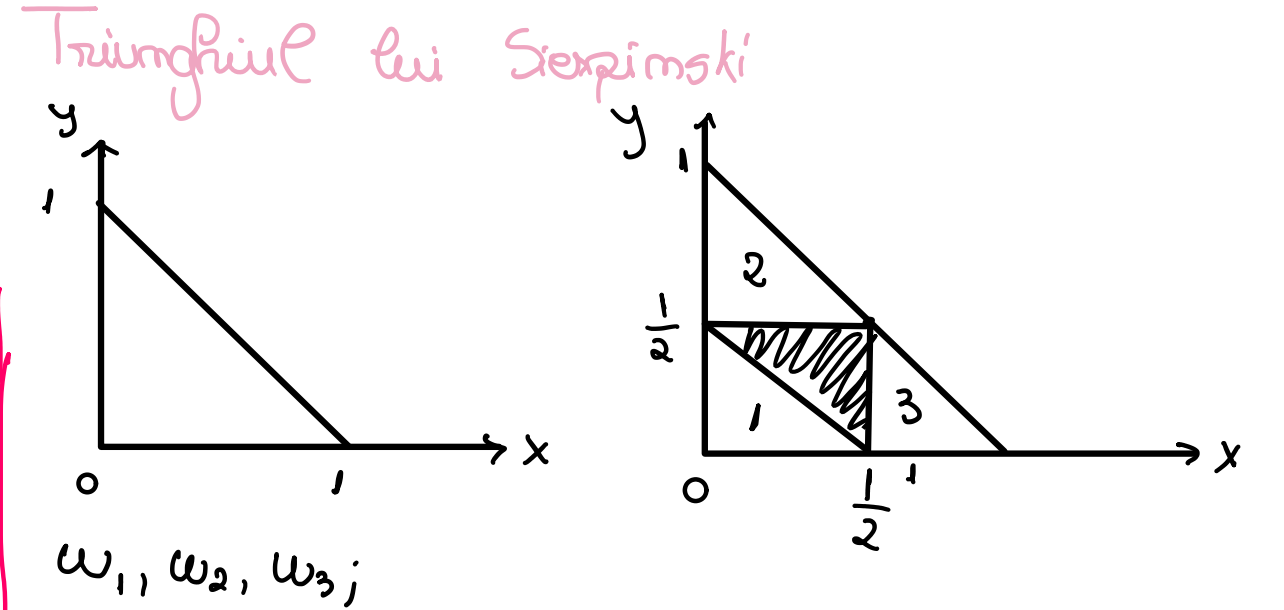
$$\omega_2(x,y)=T_{(\frac13,0)}\left(R_{\frac\pi3}\left(O_{\frac13}(x,y)\right)\right)=T_{(\frac13,0)}\begin{pmatrix}\frac12 & -\frac{\sqrt3}{2}\\ \frac{\sqrt3}{2} & \frac12\end{pmatrix}\begin{pmatrix}\frac x3\\ \frac y3\end{pmatrix}=\left(\frac x6-\frac{y\sqrt3}{6}+\frac13,\ \frac{x\sqrt3}{6}+\frac y6\right)$$

$$\omega_3(x,y)=T(\tfrac23,0)\left(R_{\frac{2\pi}{3}}\left(O_{\frac13}(x,y)\right)\right)=T_{(\frac23,0)}\begin{pmatrix}-\frac12 & -\frac{\sqrt3}{2}\\ \frac{\sqrt3}{2} & -\frac12\end{pmatrix}\begin{pmatrix}\frac x3\\ \frac y3\end{pmatrix}=\left(-\frac x6-\frac{y\sqrt3}{6}+\frac23,\ \frac{x\sqrt3}{6}-\frac y6\right)$$

$\omega_4(x,y)=T_{(\frac23,0)}\left(O_{\frac13}(x,y)\right)=\left(\frac23+\frac13x,\frac13y\right)$

$H(A)=\bigcup_{j=1}^{4}\omega_j(A)$

Triunghiul lui Sierpinski



$\omega_1,\omega_2,\omega_3$;

$\omega_1(x,y)=\left(\frac12x,\frac12y\right)=O_{\frac12}(x,y)$

$\omega_2(x,y)=T_{(\frac12,0)}\left(O_{\frac12}(x,y)\right)=\left(\frac12x+\frac12,\frac12y\right)$

$\omega_3(x,y)=T(0,\tfrac12)\left(O_{\frac12}(x,y)\right)=\left(\frac12x,\frac12+\frac12y\right)$

$H(A)=\omega_1(A)\cup\omega_2(A)\cup\omega_3(A)$

$A, H(A), H(H(A)),\dots\to A_\infty$ triunghiul lui Sierpinski

$H(A_\infty)=A_\infty$

Definirea mulțimilor Julia

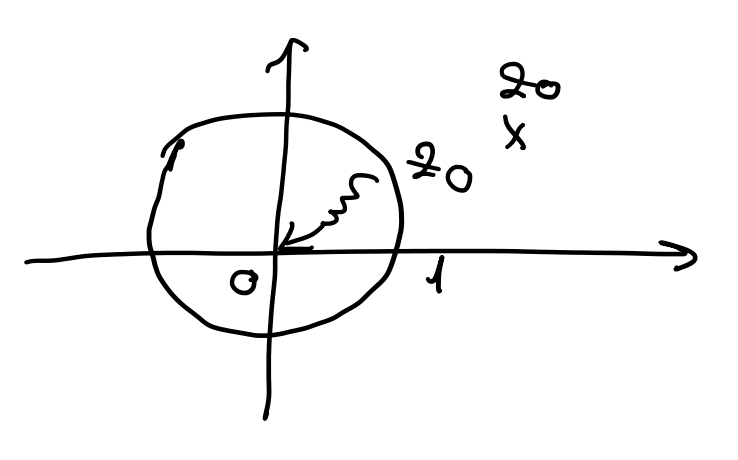
$\mathbb{C}: f:\mathbb{C}\to\mathbb{C}$

$f(z)=z^2$

$z_0\in\mathbb{C}$ arbitrar fixat

$z_1=z_0^2,\ z_2=z_0^4,\ z_3=z_0^8$

$z_n=z_0^{2^n}$



$|z_0|<1\Rightarrow z_n\to0,\ \forall n\in\mathbb{N}$

$|z_0|>1\Rightarrow z_n\to\infty,\ \forall n\in\mathbb{N}$

$|z_0|=1\Rightarrow|z_n|=1,\ \forall n\in\mathbb{N}$

$D(0,1)$ bazin de atracție

$\mathcal{C}(0,1)$

$\{z_0\mid|z_0|>1\}$ mulțime de respingere

$z_n\to0\qquad z_n\to\infty$

$f(z)=z^2$, $f(0)=0$ punct fix

$f(\infty)=\infty$

$D(0,1)$ mult. Julia plină asociată

$\mathcal{C}(0,1)\to$ mult. Julia

Fie $c\in\mathbb{C}$ fixat

$f(z)=z^2+c$

$z_0\in\mathbb{C}$ arbitrar fixat

$z_0,\ z_1=f(z_0)=z_0^2+c$

$z_2=f(z_1)=f(z_0^2+c)=(z_0+c)^2+c$

$z_n=f(z_{n-1})$

Deci $E_c=\{z_0\in\mathbb{C};|z_n|\to\infty$, pentru $n\to\infty\}$ (o mulțime Julia e frontiera mulțimii)

Teorema contracției (demonstrație)

$x_0\in X$ (arbitrar fixat)

$x_1=T(x_0),\ x_2=T(x_1)\dots$

$x_n=T(x_{n-1})$

$x_{n+1}=T(x_n),\ n\ge0$

x_n șirul aprox. succesive.

$x_n\to\xi$ și $T(\xi)=\xi$

Vom demonstra că x_n este șir Cauchy

$d(x_{n+1},x_n)=d(T(x_n),T(x_{n-1}))\le k\,d(x_n,x_{n+1})=k\,d(T(x_{n-1}),T(x_{n-2}))\le k^2d(x_{n-1},x_{n-2})\le\dots\le k^n d(x_1,x_0)\Rightarrow d(x_{n+1},x_n)\le k^n(x_1,x_0)$

$d(x_n,x_{n+p})\le d(x_n,x_{n+1})+d(x_{n+1},x_{n+2})+\dots+d(x_{n+p-1},x_{n+p})\le\frac{k^n}{1-k}d(x_0,x_1),\ \forall n\ge0\in\mathbb{N}$

$n\to\infty$ $d(x_{n+p},x_n)\to0\Rightarrow(x_n)$ șir Cauchy $\xrightarrow{X\text{-complet}}$ x_n e convergent

$\Rightarrow\exists\,\xi\in X$ a.î. $x_n\to\xi$

$p\to\infty$

$d(x_n,\xi)\le\frac{k^n}{1-k}d(x_0,x_1),\ \forall n\in\mathbb{N}$